

# Ahmed Ikram | Software Engineering

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## PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Builds production-ready systems across the full stack - React/React Native frontends, FastAPI/Flask APIs, real-time WebSocket layers, and multi-cloud infrastructure (AWS + Azure) - with 1,402 tests gating every PR across a 3-pipeline OIDC-federated CI/CD system and 118 Terraform-provisioned AWS resources with zero static credentials. Seeking a graduate Software Engineering or Full-Stack role in a team that ships production software.

## EDUCATION

### High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

### University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

## PROJECTS

### DevSync Project Management Platform with GitHub Integration | React, Tailwind, Python, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, GitHub Actions, CI/CD | Jan 2025 - Apr 2026

- React, Tailwind frontend, Flask & Gunicorn REST API over a 12-table indexed PostgreSQL schema, and a JWT-authenticated Socket.IO layer with project-scoped rooms - live task updates, notifications, and dashboard refreshes broadcast only to members viewing the affected project, with no page refresh required; GitHub activity fetched via GraphQL API to reduce latency when creating charts and reports
- Implemented three-tier RBAC (Developer / Team Leader / Admin) via role decorators on every route; JWT in HTTP-only cookies; GitHub OAuth 2.0 handled server-side with tokens never sent to the browser; bidirectional task ↔ issues/PR sync means closing a GitHub issue automatically updates the linked task in DevSync
- Production-like AWS infrastructure: ECS Fargate behind an ALB, RDS in a private subnet, and S3, CloudFront & ACM for the React SPA - frontend deployment blocked until ECS health checks pass to prevent API/UI version mismatch; 1,452 automated tests at 85%/85% coverage hard-gate every PR via 7 path-aware CI jobs (lint: ruff + ESLint, security: pip-audit + npm audit, Postgres 15 integration, Cypress E2E), OIDC federation for CI/CD with no static credentials, Docker layer caching for 70%+ faster builds, secrets injected at runtime via AWS Secrets Manager and per-task IAM roles scoped to least-privilege Dependabot + CodeQL on weekly cadence

### ATM Log Aggregation, Analysis & Diagnostics Platform | Python, FastAPI, React, PostgreSQL, Kafka, Redis, nginx, Docker, Terraform, AWS, SageMaker | Mar 2026 - June 2026

- Backend engineering lead on a 7-person Agile team delivering a diagnostics platform for NCR Atleos; independently architected a 10-service Dockerised platform, full AWS production deployment, and 3-pipeline CI/CD system - all built post-deadline on top of the team-delivered foundation
- Architected a 10-service system with health-check cascading startup: FastAPI REST API (30 endpoints, 6 routers, JWT + Redis blacklist auth), Kafka Kraft consumer with hybrid Redis+LRU deduplication and a Redis Stream DLQ, 8 Redis distributed patterns and a React frontend via multi-stage Docker build (~25MB nginx-alpine final image); deployed on AWS via Terraform (10 modules, 118 resources) - multi-AZ VPC (2 public + 2 private subnets), ECS Fargate (2 services × 2 tasks across AZs) behind an ALB, EC2 t3.medium hosting Kafka/Redis/ChromaDB/Ollama in private subnets, CloudFront serving the React SPA from S3, 8 secrets injected at container start via Secrets Manager, and 6 least-privilege IAM roles
- Built a 3-pipeline CI/CD system: CI runs ruff/mypy/bandit lint, checkov IaC compliance, and 1,402 tests (923 pytest at 82% coverage, 394 vitest, 10 Playwright E2E, 75 Terraform assertions) in parallel on every PR; CD-SHOULD-DEPLOY uses dorny/paths-filter to skip infrastructure jobs on documentation-only changes; CD applies Terraform plan/apply across all 10 modules and triggers a zero-downtime ECS rolling update on merge to main - all GitHub Actions pipelines authenticate to AWS via OIDC AssumeRole with no long-lived credentials stored anywhere

### StockLens FinTech Spending & Investment App | React Native Expo, TypeScript, Node.js, PostgreSQL | Oct 2025 - Dec 2025

- Built a full-stack mobile app in React Native/TypeScript with secure biometric authentication, real-time Firebase synchronisation, and custom AES encryption for sensitive data at rest
- Integrated a Node.js OCR pipeline parsing physical receipts into structured JSON records, querying the Alpha Vantage API to map spending to stock tickers and compute historical CAGR - covered by 78 Jest tests across OCR fallbacks, API caching, cryptography roundtrips, auth flows and more
- Engineered an ML layer combining ARIMA-based time-series forecasting for per-category spending trends and Linear Regression for forward-looking portfolio projections, fed by historical CAGR computed from Alpha Vantage data - surfaced directly in the app UI

- W3C Web Logs ETL Pipeline** | *Apache Airflow, Python, Databricks, dbt, Apache Spark, PySpark, Azure SQL, Terraform, Docker, GitHub Actions, Power BI, Grafana, Prometheus* | Feb 2026 - June 2026
- Built a serverless Databricks medallion pipeline (Bronze → Silver → Azure SQL → dbt → Power BI) ingesting 93 IIS log files (155,380 rows, 0 dropped) across 7 @dlt.expect\_or\_drop quality gates; a custom UDF handles 14- vs 18-field format detection dynamically; 4 Airflow DAGs orchestrate the full pipeline with zero data-processing logic in DAG files - layers decoupled via Airflow Datasets, no ExternalTaskSensor
  - Provisioned 30+ Azure resources across 2 Terraform configurations including the full OIDC auth chain (Azure AD app, service principal, federated credential, role assignment) via a single terraform apply; 7-workflow CI/CD pipeline with 4 parallel jobs and a rollback workflow that re-applies Terraform from the previous commit on demand — zero static credentials anywhere
  - Delivered a 7-page Power BI dashboard (156K requests, 88 countries, P95 1,149ms, 38% bot traffic, 9.7% 404 rate) from 18 dbt-generated CSV exports via a semantic contract; observability: custom Python freshness probe → 3 Grafana dashboards (23 panels) → 10 alert rules across 3 severity-tiered action groups with Slack notifications; 613 tests across 5 tiers including dual-dialect dbt CI against side-by-side PostgreSQL and SQL Server containers

**TECHNICAL SKILLS**

|                   |                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages         | Python · TypeScript · JavaScript · SQL · Bash                                                                                                                            |
| Frontend & Mobile | React · React Native · Tailwind · Vite · Chart.js · Socket.IO                                                                                                            |
| Backend & APIs    | FastAPI · Flask · Node.js · GraphQL · Kafka · Gunicorn · Airflow · dbt                                                                                                   |
| Security          | JWT · OAuth 2.0 · RBAC · AWS Secrets Manager                                                                                                                             |
| Databases         | PostgreSQL · Redis · SQLite · MySQL · MongoDB · Alembic                                                                                                                  |
| Cloud & DevOps    | AWS (ECS Fargate, EC2, S3, RDS, IAM, CloudFront, Sagemaker, Secrets Manager) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · nginx · GitHub Actions · Linux |
| Observability     | Grafana · Prometheus · StatsD · Slack (alert routing)                                                                                                                    |
| Testing           | Pytest · Vitest · Jest · Playwright · React Testing Library · Cypress · dbt                                                                                              |

**CERTIFICATES**

|                                                                            |          |
|----------------------------------------------------------------------------|----------|
| <u>AWS Academy Graduate - Machine Learning Foundations</u>                 | Feb 2025 |
| <u>AWS Academy Graduate - Cloud Foundations</u>                            | Mar 2024 |
| <u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u> | Oct 2024 |

**EXPERIENCE**

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|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Beaully Mini Market, Dundee, Scotland</b><br>Retail Assistant                                                                                         | September 2023 - Present   |
| <ul style="list-style-type: none"><li>• Troubleshoot and resolved Point-of-Sale and stock system errors quickly to minimise customer wait time</li></ul> |                            |
| <b>Zulus Peri Peri Grill, Dundee, Scotland</b><br>Front of House Worker                                                                                  | June 2021 - September 2023 |
| <ul style="list-style-type: none"><li>• Trained 5 new staff members, helping them get up to speed quickly during busy service periods</li></ul>          |                            |